

Design System

Document Version: 1.0.0

Design Philosophy: Playful Clinical (Friendly, Vibrant, Ultra-Legible & Energetic)

Target Platform: Mobile App (iOS & Android / Flutter)

1. Brand Identity & Visual Vibe

KidDose transforms the stressful, high-pressure task of pediatric medication calculation into a friendly, reassuring, and effortless experience.

Core Visual Pillars

- 1. **Playful Clinical:** Serious medical accuracy wrapped in a warm, approachable UI with rounded shapes and pastel-neon accents.
- 2. **Instant Visual Hierarchy:** Color-coded drug categories and status badges that can be recognized in a fraction of a second.
- 3. **Tactile & Motion-Rich:** Every interaction responds with satisfying spring animations and micro-haptics, giving instant tactile feedback.

2. Color System

The palette pairs high-contrast functional colors (Red Flags, Safe State, Max Caps) with cheerful accent colors for categories, avoiding cold clinical whites and dark intimidating grays.

2.1 Core Brand & Functional Tokens

Token Name	Hex Code	Visual Identity	Usage Context
Primary Blue	#2A6BFF	 Bright Electric Blue	Primary buttons, active tabs, main volume text
Safe Emerald	#00C897	 Vibrant Mint Green	Safe dosage status, verified calculations
Red Flag Crimson	#FF3B5C	 Energetic Crimson Red	Age contraindications, G6PD alerts, errors

Warning Amber	#FFB800	 Sunshine Amber	Max Dose Capped, Renal adjustment alerts
Background Warm	#F8F9FD	 Soft Ice Blue-White	Main app dashboard background
Surface Card	#FFFFFF	 Pure White	Drug result cards, modals
Text Charcoal	#1E2229	 Rich Charcoal (No pure black)	Primary headers and high-contrast text
Text Muted	#788296	 Slate Gray	Secondary captions, dosage unit labels

2.2 Category Color Accents (Vibrant Pills)

Each medical category gets its own distinct, cheerful identity to help healthcare providers navigate instantly:

- **OPD Common:** Fresh Teal (#00B8D9)
- **Antibiotics:** Electric Grape (#8E44AD)
- **ER / Resuscitation:** Coral Red (#FF5252)
- **Asthma / Respiratory:** Sky SkyBlue (#0288D1)
- **Favorites / Pinned:** Warm Gold (#FFAB00)

3. Typography & Hierarchy

The typography uses friendly, rounded sans-serif fonts with thick weights for primary figures (\$mL\$) and crisp legibility under dim hospital lighting.

- **Primary Thai Font:** *Sukhumvit Set* or *Kanit* (Semi-Bold/Bold for numbers)
- **Primary English Font:** *Plus Jakarta Sans* or *Nunito* (Rounded, modern, approachable)

[Title Display]	32pt / Bold	– Weight Input (e.g., "12.5 kg")
[Section Header]	20pt / SemiBold	– Category Name / Alerts
[Metric Highlight]	28pt / ExtraBold	– Calculated Dose (e.g., "5.0 mL")
[Body Text]	15pt / Medium	– Generic Drug Names (Thai/English)
[Caption / Tag]	12pt / Bold	– Dosage Frequency ("Q 4-6 hr")

4. Shape Language & Elevation

Avoid hard 90-degree square edges. Use soft, approachable geometry.

- **Card Corner Radius:** `20px` (Continuous smooth squircle)
- **Button / Tag Corner Radius:** `100px` (Pill shape)
- **Input Fields:** `16px` with a 2px colored border on focus.
- **Shadow Token (`Soft Float`):**
`box-shadow: 0px 8px 24px rgba(42, 107, 255, 0.08);` (Subtle blue-tinted elevation instead of harsh gray shadows).

5. Component Specifications

5.1 Single-Input Weight Bar

- **Default State:** White card with soft blue glow (`0px 8px 20px rgba(42, 107, 255, 0.12)`).
- **Focus State:** Border expands to 2.5px in `Primary Blue` with a subtle pulsing background tint.
- **Numpad Buttons:** Tactile pill-buttons with a deep press state ($0.92\times$ scale).

5.2 Drug Result Card (Interactive)

- **Normal Safe State:** White background with `Primary Blue` main volume text (5.0 mL) and green check icon.
- **Max Cap State:** Card background turns `Warm Light Amber` (`#FFF9E6`) with an Amber badge "**MAX CAP 500 mg**".
- **Red Flag Alert State:** Card background turns `Soft Crimson Tint` (`#FFEFE6`), primary volume button is replaced with a **RED WARNING** banner, and standard dose figures are blurred/locked until acknowledged.

6. Motion & Animation System

Animations in KidDose are not just decorative; they provide vital visual feedback that confirms calculations were recalculated without lagging the engine ($\leq 10\text{ ms}$).

6.1 Easing & Physics Curves

We use spring-based physics curves rather than linear transitions to create a snappy, organic feel:

```
// Flutter Custom Spring Physics Curve
static const Curve bouncySpring = Cubic(0.175, 0.885, 0.32, 1.275);
static const Curve fastOutSlowIn = Cubic(0.4, 0.0, 0.2, 1.0);
```

6.2 Key Micro-Interactions

1. Number Ticker / Roll Animation (\$mL\$ Volume Change)

- **Trigger:** User types or deletes a weight value.
- **Behavior:** The calculated \$mL\$ figure smoothly rolls up/down to the new value (like a slot machine counter) over `180ms`.
- **Implementation Hint:** `AnimatedSwitcher` with `SlideTransition` or Flutter `ImplicitlyAnimatedWidget`.

2. Red Flag "Elastic Shake" & Pulse

- **Trigger:** Patient context violates a safety rule (e.g., patient age $< 24\text{ months}$ on CPM).
- **Behavior:** The drug card triggers a fast horizontal elastic shake ($3\times$ oscillation over `250ms`), accompanied by a soft red aura pulse.
- **Haptic Sync:** Heavy impact vibration pattern (`HapticFeedback.heavyImpact()`).

3. Category Tab Pill Slide

- **Trigger:** User switches tabs (e.g., `OPD` $\rightarrow$ `ER`).

- **Behavior:** The colored indicator pill glides smoothly under the selected text with a slight overshoot stretching effect (`SpringCurve`).

4. Card Insertion / List Reordering

- **Trigger:** Toggling G6PD or Renal condition switches.
- **Behavior:** Unsafe cards smoothly scale down ($\$1.0 \rightarrow 0.95\$$) and fade out (`150ms`), while safe cards shift up smoothly (`AnimatedList`).

7. Sound & Haptic Design

To maximize confidence in busy hospital environments, tactile feedback supplements visual alerts:

Action	Haptic Type	Audio Trigger (Optional)
Typing Weight (Numpad)	Light Selection Click (<code>selectionClick</code>)	Soft Mechanical Keyboard Pop
Max Cap Reached	Medium Impact (<code>mediumImpact</code>)	Subtle Amber Chime
Red Flag Alert	Double Heavy Vibration Pattern	Soft Warning Pop
Pinning Favorite Drug	Success Vibrations (<code>vibrate</code>)	Satisfying Pop/Bubble Sound

8. Summary Checklist for UI Implementation

- ☐ **No Pure Black (`#000000`):** Always use Rich Charcoal (`#1E2229`).
- ☐ **Rounded Corners Everywhere:** Ensure all cards have at least `16px` to `20px` border radius.
- ☐ **Smooth Number Roll:** Ensure volume calculations ($\$mL\$$) use `AnimatedCounter` transitions.
- ☐ **60fps / 120fps Lock:** Ensure zero jank during animations. Logic runs in background isolates if necessary.